

System Analysis of the Project Blueprint & Research Documents

The foundational architecture of **PolyLoop** (E-Waste Plastic Identification & Reuse System) addresses the circular economy gap where 20% to 30% of total e-waste mass consists of polymeric casings that are discarded due to polymer incompatibility and halogenated additive toxicity.

The system operates across three verified scientific domains:

- 1. Thermodynamic Miscibility & Degradation Mechanics:** Macromolecular blending of post-consumer resins is constrained by positive Flory-Huggins interaction parameters ($\chi_{12} > 0$). Commingling polar Acrylonitrile Butadiene Styrene (ABS) with non-polar High-Impact Polystyrene (HIPS) causes severe micro-phase separation and over 70% loss in impact strength. Furthermore, Polycarbonate (PC) and PC-ABS blends are prone to hydrolytic chain scission if melt-extruded without precise moisture extraction below 0.02% to 0.03% (200 to 300 ppm).
- 2. Regulatory & Hazard Screening:** Brominated Flame Retardants (BFRs) such as DecaBDE, OctaBDE, and TBBPA, frequently combined with antimony trioxide (Sb_2O_3), release toxic polybrominated dioxins and furans (PBDD/Fs) when re-melted above 210°C. International standards (EU RoHS 2011/65/EU, EU POPs Regulation 2019/1021, and WEEE Directive 2012/19/EU Annex VII) strictly cap PBDEs at 500–1,000 ppm. The system predicts BFR risk across appliance vintages (e.g., pre-2006 CRT units pose a 70%–90% risk, whereas modern post-2016 laptops pose <10% risk).
- 3. Three-Tiered Identification & Closed-Loop Upcycling:** Replaces \$30,000 NIR/XRF spectroscopy systems using a 100% free software stack:
 - *Stage 1 (Object Detection):* Ultralytics YOLOv8n classifies casing types (CRT housing, PC bezel, printer door).
 - *Stage 2 (Embossed Stamp OCR):* OpenCV image filtering (Grayscale $\rightarrow$ CLAHE $\rightarrow$ Bilateral Filter $\rightarrow$ Morphological Gradient $\rightarrow$ Otsu Binarization) enhances low-contrast reliefs for EasyOCR parsing of ISO 11469 / ISO 1043 codes (>ABS<, >PS-HI<, >PC+ABS<, >ABS-FR(40)<).
 - *Stage 3 (Diagnostic Decision Wizard):* Stratification fallback using specific gravity in tap water ($\rho = 1.00 \text{ g/cm}^3$), 10% NaCl ($\rho = 1.07 \text{ g/cm}^3$), and dense brine ($\rho = 1.15 \text{ g/cm}^3$), combined with acetone and d-limonene spot solubility tests.
 - *Stage 4 (Circular Upcycling & LCA Engine):* Outputs exact 3-zone extruder temperatures (170°C–295°C), heated bed specs (85°C–130°C), and ISO 14040/14044 net carbon avoidance ($\Delta E_{\text{avoided}} = +3.02 \text{ kg CO}_2/\text{e/kg}$ for ABS, $+6.02 \text{ kg CO}_2/\text{e/kg}$ for PC) compiled into a downloadable ReportLab PDF Material Passport.

Master Agent IDE Prompt (Claude Code / Antigravity)

Copy and paste the prompt below directly into your Agent IDE.

You are an expert Principal Full-Stack AI Engineer, Computer Vision Specialist, and Sustainable Materials Scientist. Your mission is to build and deliver the complete, production-grade repository for : an end-to-end, zero-cost, software-driven E-Waste Plastic Identification, BFR Hazard Screening, and Closed-Loop Upcycling System.

- You MUST NOT use any commercial, paid, or cloud APIs (NO OpenAI, NO AWS Bedrock, NO Google Cloud Vision, NO paid vector DBs).
- The entire application MUST run locally at ZERO EXPENSE (\$0 / ₹0) using free, open-source libraries: Python 3.11+, FastAPI, OpenCV, Ultralytics YOLOv8, EasyOCR, SQLite, SQLAlchemy, React 18, Vite, Tailwind CSS, Lucide React, Recharts, and ReportLab.
- Camera input must support both a local live HTML5/WebRTC webcam feed and an image upload/sample selector.

1. Use Ultralytics YOLOv8n to identify e-waste components: ``monitor_backing``, ``pc_chassis``, ``printer_panel``, ``crt_housing``, ``keyboard_casing``. If model weights are uninitialized, provide an automated synthetic/lightweight training/export script and fallback mock heuristic inference based on aspect ratio/features.

2.

- Input: Crop ROI from user bounding box or full image.
- Step 1: Grayscale conversion: ``cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY)``
- Step 2: Contrast Limited Adaptive Histogram Equalization (CLAHE):
``cv2.createCLAHE(clipLimit=2.5, tileGridSize=(8, 8))``
- Step 3: Bilateral Denoising: ``cv2.bilateralFilter(enhanced, d=9, sigmaColor=75, sigmaSpace=75)``
- Step 4: Morphological Gradient: ``cv2.morphologyEx(blur, cv2.MORPH_GRADIENT, cv2.getStructuringElement(cv2.MORPH_RECT, (3, 3)))``
- Step 5: Otsu Binarization: ``cv2.threshold(grad, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)``

3.

- Ingest the binarized image into EasyOCR (``reader.readtext``).
- Extract standardized ISO tokens with regex: ``r'\s*([A-Z0-9|+|-|\\(|\\)|+)]\s*<'`` and secondary

fallback matching for unbracketed abbreviations: `ABS`, `HIPS`, `PS-HI`, `PC`, `PC+ABS`, `PC-ABS`, `PP`, `HDPE`, `PVC`.

- Detect flame-retardant ISO 1043-4 codes: `FR(40)` (Halogenated + Sb₂O₃), `FR(14)` / `FR(52)` (Phosphorus/Red P).

Embed a comprehensive domain knowledge dictionary:

- Density: 1.05–1.08 g/cm³, \$T

g\$: 100°C, Tap water:

Sinks slowly, 10% NaCl: Floats, Dense brine: Floats, Acetone: Swells and softens, d-Limonene: Dissolves completely into clear gel, Flame: Orange with soot webs, sweet marzipan/styrene odor.

- Density: 1.20–1.22 g/cm³, \$T

g\$: 110–125°C, Tap water: Sinks rapidly,

10% NaCl: Sinks rapidly, Dense brine: Neutral buoyancy, Acetone: Slow surface softening, d-Limonene: Inert, Flame: Yellow-orange, slight self-extinguishing behavior.

- Density: 0.90–0.96 g/cm³, Tap water: Floats.

- Density: 1.18–1.35 g/cm³, Tap water: Sinks, 10% NaCl: Sinks, Dense brine: Sinks, Beilstein copper flame test: Bright emerald green flame.

Calculate risk probability (\$0.0 \text{ to } 1.0\$) using deterministic rules:

- Appliance Typology & Vintage:

- `crt\_housing` / Pre-2006 TVs: Base risk = 0.85 (Critical Hazard: DecaBDE/OctaBDE > 2000 ppm) ->

- `laser\_printer` / Copiers (1998–2012): Base risk = 0.50 (High Hazard: Internal fuser assemblies contain TBBPA) ->

- `desktop\_chassis` (2000–2015): Base risk = 0.35 (Moderate Hazard).

- `flat\_display` / Laptops (2016–Present): Base risk = 0.08 (Low Hazard:

Organophosphorus alternatives).

- `keyboard\_mouse\_router` (2010–Present): Base risk = 0.04 (Minimal Hazard).

- ISO Code Overrides: If OCR contains `FR(40)`, set BFR Risk = 0.99 (Immediate Regulatory Rejection).

- Regulatory limits enforced: EU RoHS (< 1000 ppm / 0.1%), EU POPs (< 500 ppm), WEEE Annex VII selective extraction.

For verified non-toxic polymers, compute optimal mechanical parameters:

-
- ABS: 80°C for 3–4 hours, max moisture < 0.05% (500 ppm).
- HIPS: 70°C for 2–3 hours, max moisture < 0.05% (500 ppm).
- PC: 120°C for 4–6 hours, max moisture < 0.02% (200 ppm).
- PC-ABS: 100°C for 3–4 hours, max moisture < 0.03% (300 ppm).
-
- ABS: Feed Z1: 185–195°C, Transition Z2: 215–225°C, Die Z3: 225–235°C, Water bath: 50–60°C.
- HIPS: Feed Z1: 170–180°C, Transition Z2: 195–210°C, Die Z3: 210–225°C, Water bath: 40–50°C.
- PC: Feed Z1: 230–245°C, Transition Z2: 265–275°C, Die Z3: 280–295°C, Water bath: 70–80°C.
- PC-ABS: Feed Z1: 210–220°C, Transition Z2: 235–245°C, Die Z3: 250–265°C, Water bath: 60–70°C.
-
- ABS: Nozzle 230–245°C, Bed 95–110°C (PEI + Dimafix), Chamber 40–50°C, Fan 0–15%, Shrinkage 0.4–0.7%.
- HIPS: Nozzle 220–240°C, Bed 85–100°C, Chamber 35–45°C, Fan 20–35%, Shrinkage 0.3–0.6%.
- PC: Nozzle 280–305°C (All-Metal), Bed 115–130°C, Active Chamber 65–80°C, Fan 0%, Shrinkage 0.6–0.9%.
- PC-ABS: Nozzle 255–275°C, Bed 100–115°C, Chamber 45–60°C, Fan 10–20%, Shrinkage 0.5–0.7%.

Calculate net Global Warming Potential offset:

$$EF_{\text{recycled}} \cdot (EF_{\text{landfill}} - EF_{\text{recycling}}) + M_{\text{recycled}} \cdot EF_{\text{recycling}}$$

$$EF_{\text{recycling}} = 0.75\$$$

$$EF_{\text{credit}} = 0.05\$$$

$$EF_{\text{recycling}} = 0.75\$$$

$$EF_{\text{recycling}} + EF_{\text{landfill}} - EF_{\text{recycling}} = 0.83 \text{ kg CO}_2\text{e/kg}$$

(or Net Burden = \$0.83 \text{ kg CO}_2\text{e/kg}\$)

EF_{virgin} :

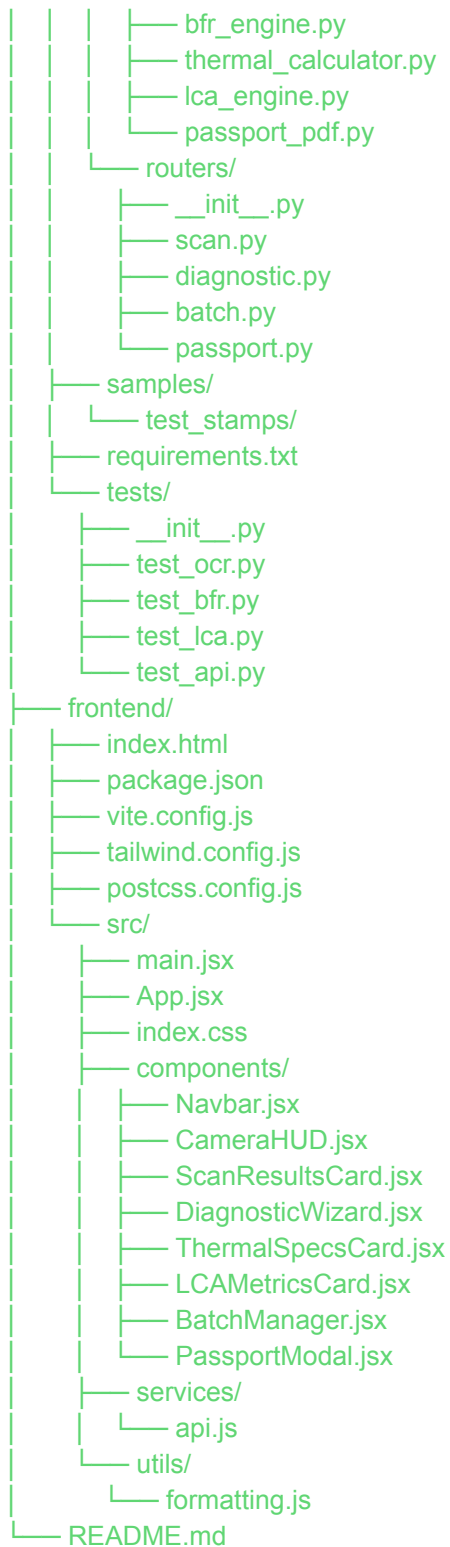
- ABS: 3.80 kg CO₂e/kg (Net Avoided: +3.02 kg CO₂e/kg; ~2.1 L crude oil)
- HIPS: 3.35 kg CO₂e/kg (Net Avoided: +2.57 kg CO₂e/kg; ~1.8 L crude oil)
- PC: 6.80 kg CO₂e/kg (Net Avoided: +6.02 kg CO₂e/kg; ~3.9 L crude oil)
- PC-ABS: 5.10 kg CO₂e/kg (Net Avoided: +4.32 kg CO₂e/kg; ~2.9 L crude oil)

Generate an official multi-section PDF document containing:

1. Header: PolyLoop Digital Material Passport & Batch Verification ID.
2. QR Code / Hash Stamp of verification payload.
3. Casing & Origin Metadata: Appliance type, estimated year, initial hazard index.
4. Identification Audit Trail: Method used (Vision YOLO / OCR Stamp / Sink-Float Wizard), detected polymer, ISO code.
5. Chemical & BFR Compliance: RoHS (Passed < 0.1%), POPs (< 500 ppm), WEEE clearance status.
6. Processing Datasheet: Moisture limits, drying profile, 3-zone extruder temps, 3D printing settings.
7. LCA Impact Statement: Net CO₂e avoided (kg), equivalent crude oil saved (L), landfill mass diverted.

Implement this exact directory hierarchy:

```
```text
polyloop/
├── .github/
│ └── workflows/
│ └── ci.yml
├── backend/
│ ├── app/
│ │ ├── __init__.py
│ │ ├── main.py
│ │ └── core/
│ │ ├── __init__.py
│ │ ├── config.py
│ │ └── constants.py
│ ├── database/
│ │ ├── __init__.py
│ │ ├── session.py
│ │ └── models.py
│ ├── schemas/
│ │ ├── __init__.py
│ │ ├── scan.py
│ │ ├── polymer.py
│ │ └── batch.py
│ └── services/
│ ├── __init__.py
│ ├── vision_service.py
│ ├── ocr_service.py
│ └── polymer_kb.py
```



### 3. BACKEND IMPLEMENTATION DETAILS

Requirements (backend/requirements.txt)

```
fastapi>=0.110.0
uvicorn[standard]>=0.28.0
pydantic>=2.6.0
sqlalchemy>=2.0.28
opencv-python-headless>=4.9.0.80
easyocr>=1.7.1
ultralytics>=8.1.0
torch>=2.2.0
torchvision>=0.17.0
numpy>=1.26.0
reportlab>=4.1.0
python-multipart>=0.0.9
pytest>=8.0.0
ruff>=0.3.0
```

## Models & Schema Specifications

- **SQLAlchemy Models (backend/app/database/models.py):**
  - ScanRecord: id (UUID/str), created\_at (datetime), casing\_type (str), vintage\_era (str), identification\_method (str: 'OCR\_STAMP' | 'SINK\_FLOAT\_WIZARD' | 'VISION\_HEURISTIC'), detected\_polymer (str: 'ABS', 'HIPS', 'PC', 'PC-ABS', etc.), confidence (float), iso\_stamp\_text (str, nullable), bfr\_risk\_score (float), rohs\_compliant (bool), status (str: 'APPROVED' | 'REJECTED'), recommended\_action (str).
  - BatchRecord: id (str), name (str), created\_at (datetime), polymer\_type (str), total\_mass\_kg (float), usable\_mass\_kg (float), rejected\_mass\_kg (float), co2\_avoided\_kg (float), oil\_saved\_liters (float), passport\_pdf\_path (str, nullable).
  - BatchItem: Relates ScanRecord to BatchRecord.

## Core Python Services

1. **polymer\_kb.py:** Complete dictionary containing the physical constants, density ranges, solvent solubility matrices, drying regimes, extrusion boundaries, and carbon emission factors described in Section 1.
2. **ocr\_service.py:**
  - Ingests image bytes / numpy array.
  - Applies the 5-step OpenCV pipeline (Grayscale -> CLAHE -> Bilateral Filter -> Morph Gradient -> Otsu).
  - Runs EasyOCR.
  - Runs regex extraction for ISO 11469 symbols (>ABS<, >PS-HI<, >PC+ABS<, >ABS-FR(40)<).

- Returns parsed polymer, confidence score, raw OCR text, and detected flame retardant codes.
- 3. **bfr\_engine.py:**
  - Computes quantitative risk score based on casing category, era of manufacture, color, and OCR tags.
  - Outputs boolean RoHS/WEEE compliance flag, risk tier (MINIMAL, LOW, MODERATE, HIGH, CRITICAL), and handling protocol.
- 4. **thermal\_calculator.py:**
  - Ingests identified polymer and returns comprehensive JSON: pre-drying temp and time, maximum permissible moisture ppm, 3-zone extruder temperatures, cooling bath temp, FDM nozzle/bed temperatures, chamber requirement, cooling fan percentage, and volumetric thermal shrinkage rate.
- 5. **lca\_engine.py:**
  - Functions `calculate_item_impact(polymer, mass_kg)` and `calculate_batch_impact(items_list)`.
  - Computes net avoided GWP ( $\text{\$kg CO}_2\text{e}$ ), crude oil displacement (liters), and coal burning offset equivalent ( $\text{\$1 kg CO}_2\text{e} \approx 0.49 \text{ kg coal}$ ).
- 6. **passport\_pdf.py:**
  - Builds clean, production-styled PDF using ReportLab SimpleDocTemplate, Table, TableStyle, and Paragraph.
  - Uses professional styling: dark slate headers, clean grid borders, color-coded badges (Green for RoHS Passed, Red for BFR Warning), and detailed engineering property tables.

## REST Endpoints (backend/app/main.py & routers)

- POST `/api/scan/casing`: Upload image -> YOLO casing classification + baseline vintage risk.
- POST `/api/scan/stamp`: Upload ROI image -> OpenCV preprocessing + EasyOCR stamp recognition + Polymer match + BFR risk check.
- POST `/api/diagnostic/wizard`: Ingest stepwise answers (density observations in water/saline, acetone spot reaction, d-limonene reaction, Beilstein flame color) -> deterministic polymer identification + confidence.
- POST `/api/batches`: Create a new recycling batch session, add verified scans, and aggregate total mass.
- GET `/api/batches/{id}/metrics`: Return live LCA and circularity charts (composition %, total kg CO<sub>2</sub>e offset).
- GET `/api/passport/{id}/pdf`: Generate and stream the downloadable ReportLab PDF Material Passport.
- GET `/api/kb/polymers`: Return all reference polymer profiles and thermal specs.

## 4. FRONTEND IMPLEMENTATION DETAILS (REACT + VITE + TAILWIND)

### Design & Theme

- Industrial Modern Dark-Mode Theme (bg-slate-950, bg-slate-900, borders border-slate-800, text text-slate-100, accents emerald-500, cyan-500, amber-500, rose-500).
- Responsive, uncluttered desktop-first layout with smooth tab navigation:
  1. **Live Camera & Stamp Scanner** (Webcam HUD + File Upload + ROI Cropper).
  2. **Guided Diagnostic Wizard** (Interactive sink-float decision tree).
  3. **Thermal & Slicing Specifications** (Extruder zones, drying times, FDM parameters).
  4. **Batch Analytics & LCA Telemetry** (Recharts graphs for mass distribution and carbon offset).
  5. **Digital Material Passport Archive** (Table of scanned batches with instant PDF download buttons).

## Key Frontend Components

1. **CameraHUD.jsx:**
  - WebRTC webcam stream with toggle to front/rear camera or pause/snapshot.
  - Fallback button: "Load Sample E-Waste Casing" with pre-bundled test images (e.g., keyboard ABS casing, old CRT monitor back, laptop bezel).
  - Targeting reticle and canvas overlay drawing bounding boxes.
2. **DiagnosticWizard.jsx:**
  - Dynamic step-by-step diagnostic questionnaire for un-stamped plastics:
    - Step 1: Tap Water Test (Floats = Polyolefin PP/PE | Sinks = Proceed).
    - Step 2: 10% NaCl Saline Test (Floats = HIPS candidate | Sinks = ABS/PC candidate).
    - Step 3: Acetone Spot Reaction (Instant tacky paste = ABS | Slow surface swell = HIPS | Resistant = PC).
    - Step 4: Optional d-Limonene Test (Dissolves into gel = HIPS | Inert = ABS/PC).
    - Step 5: Optional Beilstein Copper Wire Flame Test (Green flame = BFR/Halogen warning).
  - Generates confidence score and button to push identified item into current batch.
3. **ThermalSpecsCard.jsx:**
  - Visual 3D printer extruder graphic showing Zone 1 (Feed), Zone 2 (Transition), and Zone 3 (Die) temperature targets.
  - Drying chamber timer gauge and FDM bed/nozzle recommendation card.
4. **LCAMetricsCard.jsx:**
  - Real-time Recharts displays:
    - Bar Chart: Virgin Resin vs PolyLoop Process Carbon Footprint.
    - Doughnut/Pie Chart: Current batch composition breakdown (ABS % vs HIPS % vs PC-ABS %).
    - KPI Stat Cards: Total kg CO<sub>2</sub>e Prevented, Liters Crude Oil Equivalent Saved, Landfill Mass Diverted.
5. **PassportModal.jsx:**
  - Fullscreen preview modal showing the generated passport details with a 1-click Download PDF trigger communicating with `/api/passport/{id}/pdf`.

## 5. REPOSITORY GOVERNANCE, CI/CD & TESTING

### 1. **GitHub Actions Workflow (.github/workflows/ci.yml):**

- Triggers on push and pull requests against develop and main.
- Sets up Python 3.11, installs backend requirements, runs ruff check ., and executes pytest tests/.
- Sets up Node 20, runs npm install, and verifies npm run build.

### 2. **Automated Unit Tests (backend/tests/):**

- test\_ocr.py: Tests regex parsing against strings like >ABS<, >PS-HI<, >PC+ABS<, >ABS-FR(40)<, and noisy OCR artifacts.
- test\_bfr.py: Validates that CRT appliances and FR(40) inputs result in RoHS rejection and critical risk alerts.
- test\_lca.py: Validates carbon offset equations against standard values (e.g., 50 kg ABS yields exactly +151.0 kg CO<sub>2</sub>e avoided).
- test\_api.py: Tests FastAPI endpoint response codes and validation rules.

## 6. EXECUTION INSTRUCTIONS FOR THE AGENT

When this prompt is provided:

1. Initialize the monorepo directory tree structure immediately.
2. Create all backend files, models, routers, and services with clean, fully implemented, runnable Python code. Ensure zero placeholder stubs.
3. Build the sample test images generator in backend/samples/ using OpenCV to programmatically synthesize realistic synthetic casing stamp images with text >ABS<, >HIPS<, and >PC+ABS< for instant testing without needing external downloads.
4. Create the complete frontend application inside frontend/ with working Vite configuration, Tailwind styling, React components, and mock fallback data so the entire UI is interactive immediately even before starting the backend.
5. Provide the exact shell commands to run both the FastAPI server (uvicorn app.main:app --reload --port 8000) and the Vite client (npm run dev).

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### ### Step-by-Step Execution Guide for Claude Code / Antigravity

#### 1. **\*\*Initialize the Workspace:\*\***

Open your terminal in an empty directory and launch your agent:

```
```bash
mkdir polyloop && cd polyloop
git init
git branch -M main
git checkout -b develop
```

2. Execute Claude Code / Antigravity:

- Feed the prompt block above into the agent.
- The agent will build the files in backend/ and frontend/, configure your SQLite database, implement the OpenCV/EasyOCR pipeline, and configure the React HUD.

3. Run the Backend:

```
cd backend
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
```

4. Run the Frontend:

```
cd ../frontend
npm install
npm run dev
```

5. Verify the Demo:

- Open <http://localhost:5173> in your browser.
- Allow webcam permissions or select a sample synthetic casing.
- Inspect the live OpenCV thresholding canvas, check the automated BFR risk score, test the Sink-Float wizard, and click **Download Digital Material Passport** to receive the ReportLab-generated batch PDF.